AP BIOLOGY TOPIC 6.6

Gene Expression, Enhancers, Small RNAs & Cell Specialization

UNIT 6: GENE EXPRESSION & REGULATION (12–16% WEIGHT)

IST-2.C | IST-2.D | ENHANCER DNA LOOPING | MIRNA & SIRNA

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 6.6 (GENE EXPRESSION, ENHANCERS, SMALL RNAS & CELL SPECIALIZATION)

CED Core Vocabulary

Promoters & General Transcription Factors — Every eukaryotic gene has a proximal promoter (often containing a conserved TATA box ~25–30 bp upstream of the transcription start site).

Enhancers — are distal regulatory DNA sequences (located thousands of base pairs upstream, downstream, or even within an intron) composed of multiple control elements.

Genomic Equivalence — Virtually every somatic cell in a multicellular organism arises from the same fertilized zygote via mitosis and therefore contains the exact same complete DNA...

Sequential Expression in Embryonic Development — Cytoplasmic Determinants (e.g., maternal Bicoid mRNA): Unevenly distributed maternal mRNAs/proteins in the unfertilized egg establish early body axes...

General Transcription Factors (TFs) — bind the TATA box required for basal (low-level) transcription of all protein-coding genes and recruit RNA Polymerase II to form the transcription initiation...

Silencers — are regulatory DNA sequences that bind Repressor proteins .

Why Cells Differ (Differential Gene Expression) — Different cell types express different subsets of genes (alongside shared "housekeeping" genes like glycolysis enzymes, actin, and histones) because each cell...

Epigenetic Chromatin Modifications — Histone acetylation (ON) vs. DNA methylation (OFF).

§2. PROMOTERS, ENHANCERS, SILENCERS & TRANSCRIPTION FACTORS

Eukaryotic Initiation

- **Promoters & General Transcription Factors:**
  - Every eukaryotic gene has a proximal **promoter** (often containing a conserved **TATA box** ~25–30 bp upstream of the transcription start site).
  - **General Transcription Factors (TFs)** bind the TATA box required for basal (low-level) transcription of all protein-coding genes and recruit **RNA Polymerase II** to form the transcription initiation complex.
- **Distal Enhancers & Specific Activators (Positive Regulation):**
  - **Enhancers** are distal regulatory DNA sequences (located thousands of base pairs upstream, downstream, or even within an intron) composed of multiple control elements.
  - Specific **Activator proteins** bind enhancers; a **DNA-bending protein** folds the DNA into a hairpin loop, bringing the bound activators into contact with **Mediator proteins**, general TFs, and RNA Pol II at the promoter → **dramatically boosts transcription rate**!
- **Silencers & Repressors (Negative Regulation):**
  - **Silencers** are regulatory DNA sequences that bind **Repressor proteins**.
  - Repressors decrease or shut off transcription by blocking activator binding, preventing RNA Polymerase recruitment, or recruiting histone deacetylases to condense chromatin.

[CORE AXIOM / REMEMBER] **Enhancers + Activators = Boost Transcription (Gas Pedal via DNA Looping).**
Silencers + Repressors = Block Transcription (Brake Pedal / Negative Control).

§3. DIFFERENTIAL GENE EXPRESSION & CELL SPECIALIZATION

Same Genome, Unique Proteome

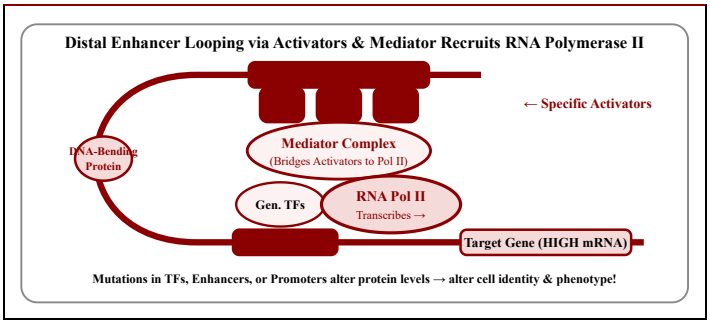
- **Genomic Equivalence:** Virtually **every somatic cell** in a multicellular organism arises from the same fertilized zygote via mitosis and therefore contains the **exact same complete DNA genome**.
- **Why Cells Differ (Differential Gene Expression):** Different cell types express **different subsets of genes** (alongside shared "housekeeping" genes like glycolysis enzymes, actin, and histones) because each cell type contains a **unique cytoplasmic combination of specific transcription factors (activators/repressors)**!
- **Sequential Expression in Embryonic Development:**
  - **Cytoplasmic Determinants (e.g., maternal *Bicoid* mRNA):** Unevenly distributed maternal mRNAs/proteins in the unfertilized egg establish early body axes (anterior-posterior).
  - **Embryonic Induction & Homeotic (*Hox*) Genes:** Signaling molecules (morphogens) secreted by neighboring embryonic cells trigger cascades of master transcription factors (such as *Hox* genes and *MyoD*) that lock in tissue identity.

[EXAMPLE & BIOLOGICAL APPLICATION] **Liver Cell vs. Lens Cell (or Muscle vs. Neuron):** Both a liver hepatocyte and an eye lens cell possess the genes for *albumin* and *crystallin*, but only liver cells make the specific activators for **albumin**, while only lens cells make the specific activators for **crystallin**!

[COMMON AP EXAM FRQ TRAP] **Differentiation ≠ Loss or Change of DNA:** Specialized cells (like neurons, muscle cells, and skin cells) differ because of **differential gene expression (which mRNAs/proteins are made)**, NOT because they have different DNA sequences!

§4. EUKARYOTIC ENHANCER LOOPING & TISSUE-SPECIFIC ACTIVATION

3D Transcription Complex



§5. POST-TRANSCRIPTIONAL CONTROL: SMALL RNAS (MIRNA & SIRNA)

RNA Interference (RNAi)

- **Three Core Mechanisms Driving Differential Gene Expression:**
  1. **Epigenetic Chromatin Modifications:** Histone acetylation (ON) vs. DNA methylation (OFF).
  2. **Cell-Specific Transcription Factors:** Activators & repressors binding enhancers/silencers.
  3. **Post-Transcriptional RNA Processing & Small RNAs:** Alternative splicing + RNA interference (RNAi).
- **Small Non-Coding RNAs (RNA Interference / RNAi):** Short (~21–23 nt) single-stranded RNA molecules that associate with protein complexes (RISC) in the cytoplasm to base-pair with target mRNAs and **prevent translation**:

SMALL RNA TYPE	COMPLEMENTARY PAIRING	MECHANISM OF POST-TRANSCRIPTIONAL SILENCING
miRNA (microRNA)	Binds **partially** complementary sequences on target mRNA	Physically **blocks ribosome translation** (or triggers deadenylation & mRNA decay); fine-tunes endogenous developmental genes.
siRNA (small interfering RNA)	Binds **perfectly (100%)** complementary mRNA sequence	Triggers immediate **enzymatic cleavage & degradation of the target mRNA** (evolved as an antiviral defense against double-stranded viral RNA).

[AP EXAM FRQ SCORING STRATEGY] **Diagnosing High mRNA with Zero Protein on the AP Exam:** If an experiment shows that a cell has **high levels of mature cytoplasmic mRNA** for Gene X, yet produces **ZERO Protein X**, look for **miRNA / siRNA post-transcriptional silencing** blocking translation or degrading the mRNA!

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 6.6 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 6.6: Gene Expression, Enhancers, Small RNAs & Cell Specialization ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 8 & 9 (pGLO Bacterial Transformation & Restriction Enzyme Analysis of DNA): Heat-shock transformation of E. coli with pGLO plasmid (bla ampicillin-resistance gene + GFP under arabinose-inducible araC promoter). Plate comparisons: −pGLO/LB (lawn, viability control), −pGLO/LB+Amp (no growth, antibiotic lethality control), +pGLO/LB+Amp (white colonies, transformation), +pGLO/LB+Amp+Ara (glowing green colonies under UV, inducible gene expression). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Transformation Efficiency = (Total colonies counted on selective plate) / (μg of plasmid DNA spread). Gel Electrophoresis Semi-Log Standard Curve: DNA fragment size (bp, $\log_{10}$ scale on y-axis) vs. migration distance (mm, linear scale on x-axis) yields a straight line for interpolating unknown fragment lengths. PCR amplification after n thermal cycles yields 2^n target DNA copies. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Molecular Disruption Case Study: Splice-site mutations in the β-globin gene (β-Thalassemia) prevent the spliceosome (snRNPs) from excising an intron or cause exon skipping, shifting the mRNA reading frame and introducing a premature nonsense stop codon (UAA/UAG/UGA). In oncology, epigenetic hypermethylation of CpG islands in tumor-suppressor promoters (e.g., BRCA1 or MLH1) or histone deacetylation condenses euchromatin into transcriptionally silent heterochromatin without altering the primary DNA sequence.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 6.6 to prove why the claim holds.

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